

sg13cmos5l_ocr_ip__bandgap

Bandgap voltage reference for IHP sg13cmos5l

Circuit description:

This is a standard bipolar bandgap reference circuit, with output trimmable by a 16-bit thermometer-encoded trim setting. The output is buffered by a follower amplifier.

Note that this is a test version. It is based on the uncharacterized PNPMPA lateral bipolar, using the existing model from sg13g2. Once this circuit has been manufactured and tested, it will be re-trimmed to the correct nominal output.

Also note: Future versions of this bandgap will replace the thermometer-encoded trim with a 4-bit binary trim input value.

Pinout:

vdd Analog power supply, nominally 3.3V

vss Analog ground

dvdd Digital power supply, nominally 1.2V

dvss Digital ground

ena Bandgap enable, digital (1.2V) domain
1 = enabled, 0 = disabled

vbg Bandgap voltage output, nominally 1.2V

trim 16-bit thermometer-encoded trim, digital (1.2V) domain. See table below

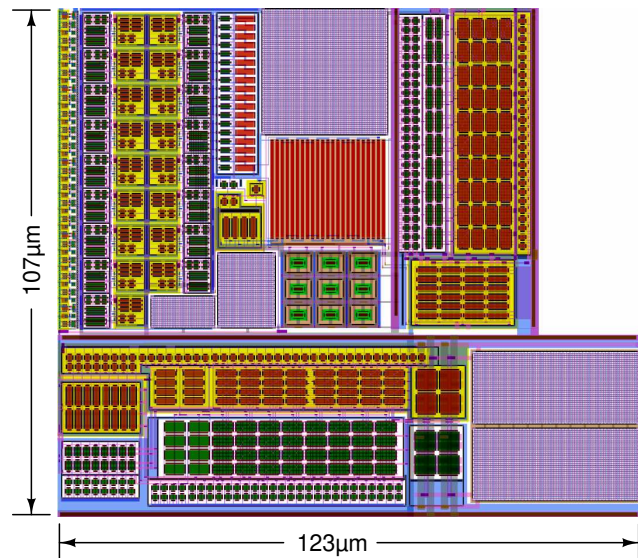
ibias1_250n Bias current for bandgap, 250nA sink. Can be provided by the bias current generator in this repository.

ibias2_1u Bias current for output buffer, 1μA sink. Can be provided by the bias current generator in this repository.

Please note that the bias current generator circuit only stabilizes the current sources with the bandgap reference, while the current sinks are self-biased. The bandgap uses only current sinks to avoid a feedback loop with the current bias generator.

trim	trim[15:0]	vbg (nom.)
0	0000000000000000	1.184V
1	0000000000000001	1.188V
2	0000000000000011	1.192V
3	0000000000000111	1.198V
4	0000000000001111	1.204V
5	0000000000011111	1.211V
6	0000000000111111	1.217V
7	0000000001111111	1.224V
8	0000000011111111	1.231V
9	0000000111111111	1.238V
10	0000001111111111	1.246V
11	0000011111111111	1.254V
12	0000111111111111	1.261V
13	0001111111111111	1.270V
14	0011111111111111	1.278V
15	0111111111111111	1.287V
16	1111111111111111	1.295V

Circuit layout*



*Note that there are different circuit layouts for the same device, with different aspect ratios.

sg13cmos5l_oed_ip__bandgap

Design: Theory of operation

This bandgap design was ported from a design for sky130 by Christoph Weiser. It is a standard voltage-mode bandgap reference, a variant of the Kuijk topology, with the output voltage taken directly from the drain of the pFET (feedback node).

The reference voltage is defined by

$$V_{bg} = V_{BE1} + (R_1 / R_2) \ln(8) V_T$$

where 8 is the current ratio between the bipolars.

The target current through the single bipolar is $I_{target} = 1.365 \mu A$

By simulation, V_{BE1} at this current is 0.750 V.

V_T at room temperature is 25.9 mV.

The nominal R_2 is therefore

$$R_2 = V_T \ln(8) / I_{target} = 39.45 k\Omega$$

The ratio of R_2 to R_1 is

$$\begin{aligned} (V_{bg} - V_{BE1}) / (V_T \ln(8)) \\ (1.25 V - 0.75 V) / 53.85 mV = 9.285 \end{aligned}$$

In this layout, $R_1 = 394.5 k\Omega$

R_2 is divided into a fixed and a variable (trimmable) component.

$$R_{2fixed} = R_1 / 10 = 39.45 k\Omega$$

$$R_{2var} (nominal) = (R_1 / 9.285) - R_{2fixed} = 3.038 k\Omega$$

The variable resistance is made with a chain of 16 resistors, with each resistor shorted by a FET when the corresponding thermometer code bit is high. By simulation, the resistance of each FET is found to be 50 Ω . The minimum resistance of $R_{2var} = 800 \Omega$ when all FETs are turned on and all resistors shorted. To make the centerpoint (half of the resistors shorted) be 3.038 k Ω , the maximum resistance should be 5.274 k Ω , and therefore each resistor should be $(5,274 / 16 \approx)$ 330 Ω .

Simulation shows the temperature response is flattest at approximately room temperature when the trim word is 0x00ff (half of the switches on, half off).

